AI-Powered Research Paper Assistant and Collaboration Platform for University Students

K D T Kariyawasam
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT22210524@my.sliit.lk

S P U Gunathilake
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT22055194@my.sliit.lk

N V Hewamanna
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT22135902@my.sliit.lk

H W S S Jayasundara
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT22553096@my.sliit.lk

Chandrasiri S. S.
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
sanji.c@sliit.lk

L I E P Weerathunga
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
ishara.w@sliit.lk

***Abstract*—Undergraduate research is supported today by tools that were never designed to work together. Students format references in one application, search literature in a second, check similarity in a third, and are allocated supervisors through largely manual processes. The resulting fragmentation costs time, permits avoidable citation and integrity errors, and leaves institutions without a consolidated view of their own research output. This paper presents the design and evaluation framework of an integrated, AI-powered research assistant and collaboration platform built from four modules behind a single interface. Module 1 operates on text entered by the student and provides a citation generator that formats references in APA and IEEE styles, a research gap analyzer based on semantic coverage analysis, a proposal generator that drafts structured proposals with retrieval-augmented generation, and a mind map generator that renders the surveyed landscape as an interactive concept graph. Module 2 matches students to supervisors and peers using Sentence-BERT embeddings and a weighted multi-factor ranking function, and classifies academic feedback with a fine-tuned BERT model. Module 3 accepts both uploaded papers and pasted text, performing multi-label topic categorisation with SciBERT, abstractive summarisation with a domain-adapted T5 model, and plagiarism analysis through combined lexical and semantic similarity. Module 4 forecasts emerging research domains with an ARIMA-Prophet ensemble and computes a four-dimensional weighted quality score. The paper describes the architecture, the design of each module, and the evaluation protocol and target thresholds for the implemented system. The platform contributes to SDG 4 and SDG 9.**

Keywords—natural language processing, recommendation systems, research integrity, academic analytics, transformer models, higher education technology

I. Introduction

Global scholarly output has grown steadily for two decades, rising from roughly 2.1 million articles in 2010 to more than 3.6 million in 2022 [1]. For an established researcher this is an opportunity. For an undergraduate undertaking a final-year project, often with limited supervision time and no formal training in research methods, it is closer to an obstacle: the literature to be surveyed has expanded far faster than the tools available to survey it.

Four difficulties recur. Citation management is the first: studies of student writing report that a substantial proportion of references contain formatting errors [2], and the problem worsens when a student must move between APA and IEEE conventions. The second is research gap identification, which demands exactly the synthesis across many papers that undergraduates are least equipped to perform [3]. The third is supervisor allocation, which in most departments remains driven by availability and prior acquaintance rather than demonstrable alignment of interests; mismatched pairings depress completion rates and output quality [4], [28]. The fourth is academic integrity: similarity checkers run after submission and return a score, offering neither an explanation nor an institutional view of where overlap is concentrating [5].

Useful tools exist for each problem, but every tool addresses one fragment. A student moves between a reference manager, a search engine, a similarity checker, a mind-mapping application and a departmental spreadsheet, none of which share data. The cost is not only time lost in switching; information generated at one stage concepts extracted from papers already read, for instance is unavailable to the stage that would benefit from it most. Institutions face the same problem, holding records of past projects and outcomes that cannot be queried as a single corpus.

This paper argues that the integration is itself the contribution. Its contributions are: (i) a four-module architecture in which citation and gap support, collaboration matching, document analysis and research analytics operate over a shared data substrate rather than as independent applications; (ii) a text-driven module combining named-entity citation generation with multi-style formatting, embedding-based gap analysis, retrieval-grounded proposal drafting and knowledge-graph mind mapping; (iii) a recommendation engine that replaces keyword-based supervisor allocation with semantic matching under an explicit, configurable multi-factor policy, together with a feedback sentiment model designed for the register of academic critique; (iv) a document analysis module that accepts both uploaded papers and pasted text and performs topic categorisation, summarisation and corpus-level plagiarism analysis; and (v) an analytics module providing ensemble trend forecasting and multi-dimensional quality scoring, with an evaluation protocol whose thresholds were fixed in advance of implementation.

Section II reviews prior work and states the resulting gaps. Section III describes the architecture and algorithmic design. Section IV sets out the evaluation protocol and discusses the trade-offs it is intended to expose. Section V concludes.

II. Background and Literature Survey

A. Citation Generation, Gap Analysis and Mind Mapping

Iqbal et al. [8] survey a decade of in-text citation analysis and observe that the field has concentrated on classifying citation contexts and purposes rather than producing correctly formatted references. CERMINE [9] extracts structured metadata using rules combined with supervised models and performs well on titles, authors and abstracts, yet does not render citations in a target style. Neither addresses the task a student faces: turning a raw reference string into a correctly formatted entry.

Automated gap analysis is less mature. ResearchAgent [10] uses large language models to propose research ideas iteratively, but generates ideas rather than mapping what a corpus does not cover; Si et al. [11] found model-generated ideas judged more novel than human ideas but less feasible, which argues for grounding both analysis and generation in the evidence actually supplied. On proposal drafting, ProposaGen [12] shows that fine-tuned LLaMA models can produce customised proposals, though for general projects rather than academic research with its particular structural conventions. Mind map generation has been demonstrated from summarised text [15], knowledge-graph prompting [16] and collaborative learning [17], but always as a standalone artefact disconnected from the gap analysis that would make it informative.

B. Collaboration and Recommendation

Chen and Chen [4] survey academic recommendation systems and note that most recommend papers rather than people, and that supervisor allocation is rarely modelled. Where matching is attempted it rests on keyword overlap, which fails precisely when a student and a supervisor describe the same problem in different vocabulary. Sentence-BERT [18] addresses that failure by producing fixed-length representations whose cosine similarity tracks semantic rather than lexical agreement. Johnson et al. [28] model supervisor effectiveness from outcome data, but treat effectiveness as an exercise separate from allocation. Sentiment analysis of academic feedback remains underexplored: general-purpose classifiers built on BERT [19] tend to label supervisor comments as negative when they are in fact constructive, because the register of academic critique differs sharply from the product-review corpora on which such classifiers are trained. This motivates both domain fine-tuning and a label set separating constructive criticism from genuine negativity.

C. Document Analysis and Integrity

Institutional repositories categorise research outputs with manually assigned keywords that are inconsistent across departments and years. Domain-pretrained encoders have improved classification of scientific text substantially: SciBERT [20] outperforms general-purpose BERT and suits multi-label assignment of thematic categories, while unsupervised approaches such as BERTopic [25] surface emergent themes a fixed taxonomy misses. For summarisation, T5 [21] and BART [22] produce abstractive summaries that read more naturally than extractive alternatives, though both require domain adaptation for scientific abstracts; ROUGE [31] remains the standard measure.

In plagiarism detection, Foltýnek et al. [5] find across more than two hundred studies that verbatim copying is handled well while paraphrase remains difficult, and that combining lexical, syntactic and semantic measures helps; Gupta and Vani [13] and Arabi and Akbari [14] both confirm that hybrid similarity outperforms purely lexical matching. Thompson et al. [6] further show that text mining can reveal plagiarism trends across a corpus rather than within a single document an institutional view that per-document similarity checkers do not provide, and one that is only available when categorisation and similarity analysis operate over the same corpus.

D. Research Analytics

Forecasting research trends is a time-series problem with irregular sampling and structural breaks. ARIMA [27] captures linear dependence through autoregression, differencing and moving-average terms, while Prophet [26] decomposes a series into trend, seasonality and event components and tolerates the missing observations common in academic data; ensembles of the two generally outperform either alone. Research quality assessment has relied on citation counts, whose long lag makes them meaningless for a student project, and on rubrics applied manually and inconsistently. Automating a rubric requires a scoring function whose dimensions and weights are explicit and auditable rather than learned opaquely.

E. Summary of Identified Gaps

Six gaps follow from this review. No platform unifies citation support, collaboration matching, document analysis and research analytics (G1); reference managers still require manual entry while metadata extractors stop short of formatted output (G2); idea generation is addressed but systematic corpus coverage analysis and its visualisation are not (G3); supervisor allocation remains manual or keyword-based, and no sentiment model is tuned to academic critique (G4); similarity checking reports a per-document score rather than an institutional overlap trend (G5); and no consolidated corpus supports trend forecasting or transparent quality assessment (G6). G1 is in large part the cause of the remainder, since a fragmented toolchain gives no component access to what another has produced.

III. Methodology

The platform follows Design Science Research [29], in iterative cycles of problem identification, artefact design, development and evaluation, with two-week agile sprints and each module evaluated independently before integration. Fig. 1 shows the architecture. Document ingestion is deliberately centralised: This keeps file handling, and the security surface that comes with it, in one place, and lets Module 1 remain a lightweight interactive tool. A REST integration bus separates the layers, which is what allows a module to consume another's output without depending on its internals.

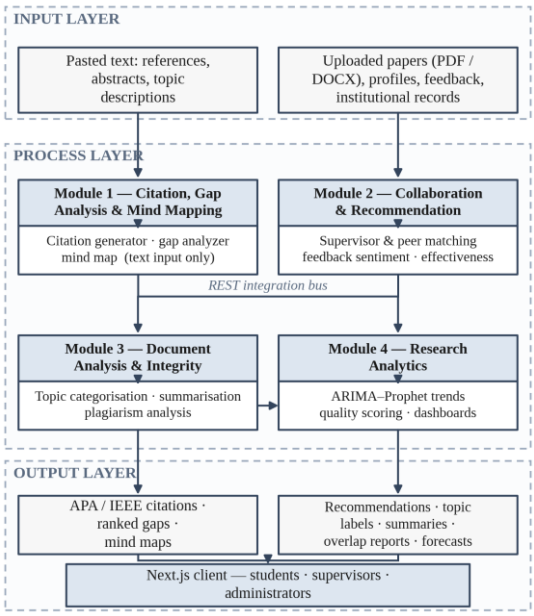


Fig. 1. Three-layer architecture of the platform.

A. Module 1: Citation, Gap Analysis, Proposal Drafting and Mind Mapping

This module, shown in Fig. 2, is the student's entry point and accepts text only pasted reference strings, abstracts or a topic description. It comprises four components sharing one input.

The citation generator uses a spaCy named entity recognition model, fine-tuned on annotated citation strings, to label authors, title, venue, year, volume, page range and DOI. A template engine maps those entities onto APA 7th edition and IEEE reference formats. Separating extraction from formatting means an additional style requires a new template rather than a retrained model, and because the input is plain text the student can correct a misparsed field directly.

The research gap analyzer encodes the supplied abstracts with Sentence-BERT [18] and clusters them; BERTopic [25] and latent Dirichlet allocation extract the dominant themes; and a coverage matrix maps themes against documents, so that themes represented by few papers, or regions of embedding space with low density, become candidate gaps. Candidates are ranked by recency, citation frequency and distance from well-explored clusters, separating a genuinely open gap from a merely obscure one.

The proposal generator is a retrieval-augmented generator [24]. A LoRA adaptation [23] of an open-weight instruction model is fine-tuned on curated academic proposals, and at inference the retrieval component fetches passages from the corpus the student supplied, constraining the draft to reference literature that exists rather than inventing it. Output is structured by section problem statement, objectives, methodology and expected outcomes each independently regenerable, and every draft carries an explicit notice that it requires review. The gap ranking feeds the generator, so the drafted problem statement addresses a gap the analyzer has identified.

The mind map generator extracts concepts with KeyBERT and TextRank, assembles them into a NetworkX graph whose edges are weighted by co-occurrence frequency, and renders the result as an interactive D3.js map. Because it draws on the same embeddings as the gap analyzer, candidate gaps are highlighted directly on the map, so the student sees unexplored regions in context rather than as a separate ranked list.

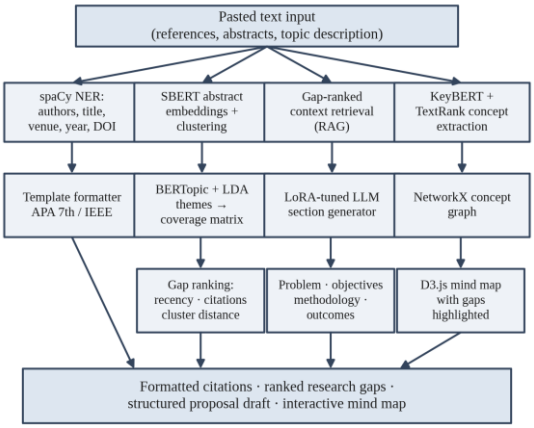


Fig. 2. Module 1 pipeline: citation generation, gap analysis and mind mapping from text input.

B. Module 2: Collaboration and Recommendation

The engine, shown in Fig. 3, represents students and supervisors in a shared 768-dimensional semantic space. A supervisor profile is a composite embedding of recent publication abstracts, stated interests and project summaries, weighted by recency and citation impact; a student is embedded from the proposal text. Cosine similarity between the two yields an interest alignment score.

Similarity alone is a poor allocation rule, because the best-aligned supervisor may be unavailable. The final score combines interest similarity (0.45), methodological alignment (0.25), workload balance (0.15) and compatibility indicators drawn from historical pairings (0.15), with weights configurable per department. The system returns the top K supervisors with an explanation naming the factors behind each ranking, since an unexplained allocation is unlikely to be trusted. Peer matching inverts the objective, scoring complementarity rather than similarity: skills a student lacks and a collaborator possesses. Content-based complementarity is combined with collaborative filtering over implicit signals such as co-enrolment, falling back to content-based scoring when data is sparse.

Feedback sentiment is classified by a BERT model [19] fine-tuned with LoRA [23] on annotated academic feedback under four labels positive, constructive-critical, negative and neutral. The constructive-critical category is included precisely because it is what a general-purpose classifier collapses into negative. The model also assigns aspect-level scores for methodology, writing quality, literature coverage, originality and data analysis, and tracks sentiment across submission cycles so improvement trajectories become visible. Supervisor effectiveness is computed from feedback constructiveness, student completion and outcome data, research productivity and response timeliness, and enters the ranking function as one input among several rather than as a published league table.

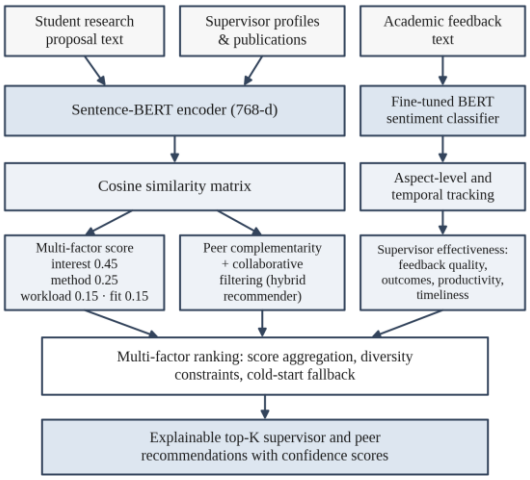


Fig. 3. Module 2 pipeline: supervisor and peer recommendation with feedback sentiment analysis.

C. Module 3: Document Analysis and Integrity

This module, shown in Fig. 4, is the only component that ingests files. It accepts uploaded papers in PDF and DOCX form as well as pasted text, extracting content with PyMuPDF and pdfplumber and applying tokenisation, lemmatisation and academic stop-word removal while preserving domain vocabulary. The three analyses that follow share this normalised representation.

Topic categorisation uses SciBERT [20] with a multi-label sigmoid head, so that a project spanning machine learning and cybersecurity receives both labels rather than being forced into one. Categories cover the departmental taxonomy, from artificial intelligence and machine learning through networking and cybersecurity to software engineering. BERTopic [25] runs alongside to surface themes outside that taxonomy, and each classification carries a confidence score, with low-confidence cases routed for human review rather than accepted silently.

Summarisation uses a T5 model [21] fine-tuned on paper–abstract pairs, with BART [22] for longer documents. Users select summary length and focus, and output is validated with ROUGE [31] and BERTScore before storage.

Plagiarism analysis combines TF-IDF vectors, which capture verbatim overlap, with SBERT embeddings, which capture paraphrase. Pairwise cosine similarity is computed against the accumulated corpus within and across cohorts, and pairs exceeding configurable thresholds are reported with the specific overlapping passages rather than a bare percentage. Because the analysis runs over the categorised corpus, the output is also a trend view: heatmaps showing which topic areas carry the highest overlap and which are saturating, with alerts raised to coordinators.

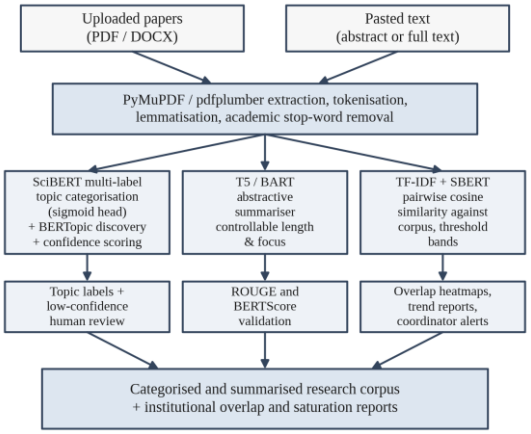


Fig. 4. Module 3 pipeline: topic categorisation, summarisation and plagiarism analysis from uploads or text.

D. Module 4: Research Analytics

Module 4 consumes the categorised corpus produced together with historical institutional records, and provides two analyses, shown in Fig. 5.

Trend forecasting aggregates records into monthly and quarterly series of submissions, topic frequencies and publication counts. ARIMA parameters are selected by the Akaike information criterion with stationarity confirmed by the augmented Dickey–Fuller test, and seasonal variants explored for annual cycles [27]; Prophet is fitted in parallel [26]. Forecasts combine in a weighted ensemble whose weights are set by relative validation performance, letting the linear model dominate stable series while Prophet handles those with structural breaks.

Quality scoring is deliberately multi-dimensional and transparent. Table I gives the four dimensions, their weights and how each is assessed. The composite score Q is

$$Q = 0.30\,S_o + 0.25\,S_c + 0.25\,S_m + 0.20\,S_p \qquad (1)$$

where each component is normalised to the range 0–100. The weights were set through a Delphi consultation and remain configurable, because they encode an institutional judgement rather than a universal one. Both analyses surface in role-based D3.js dashboards student, supervisor and administrator views with Power BI export and WebSocket updates.

TABLE I. RESEARCH QUALITY SCORING DIMENSIONS

Dimension	Weight	Assessment method
Originality	0.30	Cosine similarity and TF-IDF against the existing corpus
Citation impact	0.25	Field-normalised citation counts of cited references
Methodological rigour	0.25	NLP analysis of the methodology section for completeness
Presentation clarity	0.20	Flesch–Kincaid readability, structure and coherence

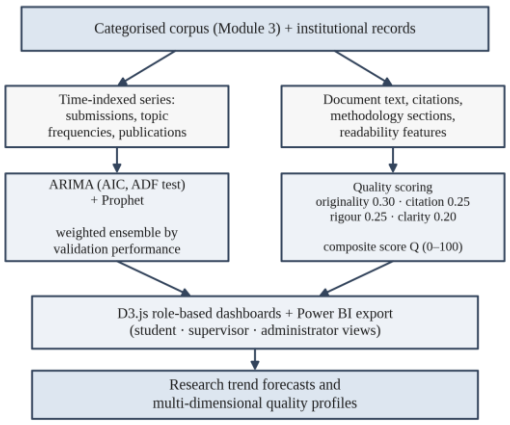


Fig. 5. Module 4 pipeline: trend forecasting and multi-dimensional quality scoring.

E. Implementation and Integration

The four modules are implemented as independent Python services behind FastAPI, coordinated by a Node.js and Express gateway with a Next.js client. Structured metadata is held in PostgreSQL, and embeddings and documents in MongoDB and object storage. Modules communicate only through versioned REST contracts, allowing each to be developed and scaled independently: inference-heavy services run on GPU instances while ingestion runs on general-purpose compute. Transport uses TLS 1.3 and data at rest AES-256, access is controlled by JSON Web Tokens under role-based authorisation, records are pseudonymised, and ethical clearance was sought from the institutional review board.

IV. Results and Discussion

A. Evaluation Protocol

Each module is evaluated independently against thresholds fixed before implementation, and then jointly in an end-to-end user study. Component evaluation uses held-out annotated data: the citation generator on annotated reference strings from IEEE Xplore, ACM and arXiv; the topic classifier on institutional projects labelled by two annotators; the summariser against author-written abstracts; the plagiarism analyzer on a corpus seeded with known paraphrased and verbatim overlaps; and the forecasting ensemble on a chronological hold-out. Gap analysis, proposal drafts and mind map quality are rated by domain experts, since no ground truth exists. The user study recruits undergraduate students and their supervisors, uses a counterbalanced design, and measures task completion time against their current toolchain alongside System Usability Scale scores [30]. Table II lists the metrics and thresholds adopted.

TABLE II. Evaluation Metrics and Target Thresholds

Module	Metric	Target
M1	Citation format accuracy	≥ 90%
M1	Entity extraction F1-score	≥ 0.85
M1	Gap precision (expert rated)	≥ 75%
M1	Proposal drafting time saved	60–70%
M1	Mind map concept relevance	≥ 80%
M2	Supervisor matching accuracy	≥ 85%
M2	Feedback sentiment F1-score	≥ 0.88
M3	Topic categorisation accuracy	≥ 92%
M3	ROUGE-L gain over generic model	≥ 40%
M3	Plagiarism detection precision	≥ 90%
M4	Trend accuracy (1 − MAPE)	≥ 78%
M4	Quality agreement (Cohen's κ)	≥ 0.70
M4	Dashboard usability (SUS)	≥ 70

B. Discussion

The thresholds are not arbitrary. The 90% citation target reflects the point at which checking a reference list becomes spot-checking rather than re-entry, where the time saving accrues. The 0.88 F1 target for feedback sentiment exceeds the supervisor matching target because the cost profiles differ: a mediocre recommendation is corrected by the student ignoring it, whereas misclassified feedback propagates into the effectiveness score and into future rankings. Plagiarism precision is set above recall, since a false accusation harms a student more than a missed overlap a supervisor would catch.

Two design decisions deserve comment. Restricting Module 1 to text input is deliberate rather than a parser limitation: centralising file ingestion in Module 3 keeps document handling and its security surface in one service, and lets the citation generator respond immediately to a pasted reference. Whether students find that division natural is a question the user study will answer. Grounding the proposal generator in retrieved passages reduces fabricated citations but constrains the draft to the corpus supplied, so for a thinly represented topic it will be conservative. The multi-factor matching function improves on pure similarity, but its weights encode a policy judgement about how far workload should override interest alignment, which is why they are departmental configuration rather than fixed in code.

The strongest argument for integration appears in the data flow rather than any single component. Module 1 uses the same embedding model that Module 2 uses for matching and Module 3 for similarity analysis, so a mind map can be annotated with saturation data from Module 3, and Module 4 forecasts over the corpus Module 3 has categorised. None of this is available to a toolchain of independent applications, and these are the outcomes the end-to-end study measures.

C. Limitations

This paper reports architecture, algorithmic design and evaluation protocol; the figures in Table II are the targets against which the implemented system is being assessed, not measured results. Three limitations should be stated plainly. First, the institutional corpus is drawn from a single university, so the topic taxonomy and forecasting models will reflect its disciplinary composition, and external validity requires replication elsewhere. Second, supervisor effectiveness scoring aggregates outcome data itself shaped by which students a supervisor is assigned a confound the composite score mitigates but does not remove, and the reason the score is used internally rather than published. Third, the language models used for drafting and summarisation inherit the biases of their pretraining corpora, which are predominantly English.

V. Conclusion

This paper has presented the design of an integrated, AI-powered research assistant and collaboration platform for

university students, built around four modules that share a common data substrate. Module 1 generates formatted citations, analyses research gaps, drafts grounded proposals and renders mind maps from text the student enters; Module 2 recommends supervisors and peers and analyses academic feedback; Module 3 ingests uploaded papers and text to perform topic categorisation, summarisation and plagiarism analysis; and Module 4 forecasts research trends and scores research quality across four weighted dimensions. The central claim is that fragmentation of the current toolchain is itself the source of much of the difficulty students face, and that the value of consolidation lies in what each stage passes to the next. Work in progress is completing the component evaluations and the user study; subsequent work will report measured performance against Table II and extend the citation generator to further reference styles. The platform contributes to SDG 4 by making research support available independently of institutional resourcing, and to SDG 9 by strengthening academic research infrastructure [7].

ACKNOWLEDGMENT

The authors thank the academic staff of the Department of Information Technology, Sri Lanka Institute of Information Technology, for their guidance and support.

REFERENCES

[1] National Science Foundation, "Publication output: U.S. trends and international comparisons," NSF, Arlington, VA, USA, 2024.

[2] A. Kumar and R. Patel, "Natural language processing for academic research paper analysis," J. Inf. Sci., vol. 47, no. 3, pp. 312–328, 2021.

[3] Y. Zhou, X. Wang, and Z. Liu, "Automated citation analysis and gap identification in literature reviews," Scientometrics, vol. 128, no. 4, pp. 1875–1892, 2023.

[4] M. Chen and J. Chen, "A comprehensive survey of recommendation systems in academic environments," IEEE Trans. Educ., vol. 64, no. 2, pp. 125–142, 2021.

[5] T. Foltýnek, N. Meuschke, and B. Gipp, "Academic plagiarism detection: A systematic literature review," ACM Comput. Surv., vol. 52, no. 6, pp. 1–42, 2019.

[6] E. Thompson, M. Anderson, and S. Brown, "Plagiarism detection and trend analysis using text mining," ACM Trans. Internet Technol., vol. 20, no. 3, pp. 1–22, 2020.

[7] United Nations, "The sustainable development goals report 2023," United Nations Publications, New York, NY, USA, 2023.

[8] S. Iqbal, S. U. Hassan, N. R. Aljohani, S. Alelyani, and R. Nawaz, "A decade of in-text citation analysis based on natural language processing and machine learning techniques," Scientometrics, vol. 126, no. 8, pp. 6551–6599, 2021.

[9] D. Tkaczyk, P. Szostek, M. Fedoryszak, P. J. Dendek, and Ł. Bolikowski, "CERMINE: Automatic extraction of structured metadata from scientific literature," Int. J. Doc. Anal. Recognit., vol. 18, no. 4, pp. 317–335, 2015.

[10] J. Baek, S. K. Jauhar, S. Cucerzan, and S. Y. Hwang, "ResearchAgent: Iterative research idea generation over scientific literature with large language models," in Proc. NAACL-HLT, 2025.

[11] C. Si, D. Yang, and T. Hashimoto, "Can LLMs generate novel research ideas? A large-scale human study with 100+ NLP researchers," 2024, arXiv:2409.04109.

[12] P. Saranath, D. Janani, and S. Hitesh, "ProposaGen: AI-driven automated proposal generation system," in Proc. Int. Conf. Next Gener. Comput. Technol., 2025.

[13] D. Gupta and K. Vani, "Using natural language processing techniques and fuzzy-semantic similarity for automatic external plagiarism detection," in Proc. Int. Conf. Adv. Comput., Commun. Inform., 2014.

[14] H. Arabi and M. Akbari, "Improving plagiarism detection in text document using hybrid weighted similarity," Expert Syst. Appl., vol. 207, pp. 1–14, 2022.

[15] A. Kulkarni, H. Shah, and L. D'Mello, "Flowchart generation and mind map creation using extracted summarized text," in Proc. Int. Conf. Recent Trends Mach. Learn., IoT, Smart Cities Appl., 2023.

[16] Y. Wen, Z. Wang, and J. Sun, "MindMap: Knowledge graph prompting sparks graph of thoughts in large language models," in Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics, 2024.

[17] Y. Jia, X. E. Wang, Z. P. T. Sin, C. Li, and P. H. F. Ng, "Knowledge-graph-driven mind mapping for immersive collaborative learning," IEEE Trans. Learn. Technol., vol. 17, pp. 1524–1537, 2024.

[18] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. EMNLP, 2019, pp. 3982–3992.

[19] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, 2019, pp. 4171–4186.

[20] I. Beltagy, K. Lo, and A. Cohan, "SciBERT: A pretrained language model for scientific text," in Proc. EMNLP-IJCNLP, 2019, pp. 3615–3620.

[21] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer," J. Mach. Learn. Res., vol. 21, no. 140, pp. 1–67, 2020.

[22] M. Lewis et al., "BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension," in Proc. ACL, 2020, pp. 7871–7880.

[23] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in Proc. Int. Conf. Learn. Represent., 2022.

[24] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Proc. Adv. Neural Inf. Process. Syst., 2020, pp. 9459–9474.

[25] M. Grootendorst, "BERTopic: Neural topic modeling with a class-based TF-IDF procedure," 2022, arXiv:2203.05794.

[26] S. J. Taylor and B. Letham, "Forecasting at scale," Amer. Statistician, vol. 72, no. 1, pp. 37–45, 2018.

[27] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, Time Series Analysis: Forecasting and Control, 5th ed. Hoboken, NJ, USA: Wiley, 2015.

[28] L. Johnson, K. Miller, and R. Davis, "Machine learning approaches for supervisor effectiveness evaluation in higher education," Comput. Educ., vol. 165, pp. 1–15, 2021.

[29] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," MIS Quart., vol. 28, no. 1, pp. 75–105, 2004.

[30] J. Brooke, "SUS: A quick and dirty usability scale," in Usability Evaluation in Industry. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[31] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in Proc. ACL Workshop Text Summarization Branches Out, 2004, pp. 74–81.